Advanced Programming Practice · 21CSC203P

Unit 5 · Session 5 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 5 : SLO â€“ 1 | Algebraic Simplification Using SymPy

I. Questions & Answers

1. Simplify (xÂ² â^' 1)/(x â^' 1) â€" simplify((x**2 â^' 1)/(x â^' 1)) â†' x + 1 (the factor x â^' 1 cancels; valid for x â‰ 1).

2. What simplify() does â€" applies a battery of heuristics (algebraic, trigonometric, logarithmic rules) to rewrite an expression in a simpler, more compact equivalent form.

3. Expand (x+1)(xâ^'1) â€" expand((x + 1)*(x â^' 1)) â†' x**2 â^' 1 (difference of squares).

II. Code

- from sympy import symbols, simplify, expand
- x = symbols('x')
- print(simplify((x**2 - 1)/(x - 1))) # x + 1
- print(expand((x + 1)*(x - 1))) # x**2 - 1